
NUMERICAL TECHNIQUES AND DIFFERENTIAL EQUATIONS

Course Code: MAC21

Credits: 2:1:1

Pre-requisites: -

Contact Hours: 28L+14T+14P

Course Coordinators: Dr. Monica Anand & Dr. B. Azghar Pasha

Course Content

Unit I

Series Expansion of Functions & their Applications: Taylor's series and Maclaurin's series expansion of one variable (without proof) and its applications to solve algebraic and transcendental equations - Newton-Raphson method.

Taylor's series and Maclaurin's series expansion of two variables (without proof) and its applications to solve the system of nonlinear equations - Newton-Raphson method. Maxima and minima of functions of two variables, Lagrange's method of undetermined multipliers.

- Pedagogy/Course delivery tools: Chalk and talk, Power Point Presentation, Videos
- Links: <https://nptel.ac.in/courses/111/106/111106101/>
<https://nptel.ac.in/courses/111/105/111105121/>
- Impartus recording: <https://a.impartus.com/ilc/#/course/119640/593>
<https://a.impartus.com/ilc/#/course/59742/295>
<https://a.impartus.com/ilc/#/course/619570/1030>

Unit II

First order and first degree differential equations: Applications of first order and first degree differential equations to solve simple electric circuit problems, Newton's law of cooling and orthogonal trajectories.

Numerical solution of first order ODE: Taylor's series method, Euler's & modified Euler's method, fourth order Runge-Kutta method.

- Pedagogy/Course delivery tools: Chalk and talk, Power Point Presentation, Videos
- Links: <https://nptel.ac.in/courses/111/106/111106100/>
<https://nptel.ac.in/courses/111/106/111106101/>
- Impartus recording: <https://a.impartus.com/ilc/#/course/119640/593>
<https://a.impartus.com/ilc/#/course/59742/295>
<https://a.impartus.com/ilc/#/course/619570/1030>

Unit III

Linear differential equations of higher Order: Linear differential equations of higher order with constant coefficients. Cauchy's and Legendre's linear differential equations, Method of variation of parameters – Engineering applications.

- Pedagogy/Course delivery tools: Chalk and talk, Power Point Presentation, Videos
- Links: <https://nptel.ac.in/courses/111/105/111105121/>
- Impartus recording: <https://a.impartus.com/ilc/#/course/119640/593>
<https://a.impartus.com/ilc/#/course/59742/295>

Unit IV

Interpolation: Forward and Backward differences, Interpolation, Newton-Gregory Forward and Backward Interpolation, Lagrange's interpolation and Newton's divided difference interpolation techniques (no proof).

Numerical Differentiation: Derivatives using Newton-Gregory forward and backward interpolation formula.

Numerical Integration: Newton-Cotes quadrature formula: Trapezoidal, Simpson's 1/3rd and Simpson's 3/8th rule (no proof).

- Pedagogy/Course delivery tools: Chalk and talk, Power Point Presentation, Videos
- Links: <https://nptel.ac.in/courses/111/106/111106101/>
- Impartus recording: <https://a.impartus.com/ilc/#/course/96127/452>
<https://a.impartus.com/ilc/#/course/132243/636>

Unit V

Linear Algebra: Elementary transformations on a matrix, Echelon form & rank of a matrix, Consistency of system of linear equations. Gauss elimination, Gauss – Seidel method to solve system of linear equations. Eigen values and eigen vectors of a matrix, Rayleigh power method to determine the dominant eigen value of a matrix, diagonalization of square matrices. Solution of system of ODEs by matrix method.

- Pedagogy/Course delivery tools: Chalk and talk, Power Point Presentation, Videos
- Links: <https://nptel.ac.in/courses/111/108/111108066/>
<https://nptel.ac.in/courses/111/105/111105035/>
- Impartus recording: <https://a.impartus.com/ilc/#/course/619570/1030>

Text Book:

1. **George B. Thomas, Maurice D. Weir, Joel R. Hass** - Thomas' Calculus, Pearson, 13th edition, 2014.
2. **B.S. Grewal** – Higher Engineering Mathematics, Khanna Publishers, 44th edition, 2017.

Reference Books:

1. **Erwin Kreyszig** - Advanced Engineering Mathematics, Wiley publication, 10th edition, 2015.
2. **Peter V. O' Neil** - Advanced Engineering Mathematics, Thomson Brooks/ Cole, 7th edition, 2011.
3. **Glyn James & Phil Dyke** - Advanced Modern Engineering Mathematics, Pearson Education, 5th edition, 2018.
4. **Srimanta Pal & Subobh C Bhunia** - Engineering Mathematics, Oxford University Press, 3rd Reprint, 2016.

Course Outcomes (COs):

At the end of the course the student will be able to:

1. Determine extreme values, roots of algebraic & transcendental equations and solution of non-linear system of equations. (PO-1, PO-2)
2. Solve first order differential equations analytically and numerically. (PO-1, PO-2)
3. Solve higher order linear differential equations with constant and variable coefficients. (PO-1, PO-2)
4. Interpolate, differentiate and integrate a given set of tabulated data (PO-1, PO-2)
5. Solve system of linear equations and ordinary differential equations using matrices. (PO-1, PO-2)

List of Programs:

1. Analyzing extreme values of functions of two variables
2. Finding the roots of algebraic and transcendental equations using Newton-Raphson method
3. Solution of system of non-linear equations using Newton-Raphson method
4. Taylor's series and Euler's method to solve ODE's of first order and first degree
5. Modified Euler's method & Runge-Kutta method of 4th order to solve first order and first degree ODE's
6. Solution of higher order ODE's
7. Interpolation/Extrapolation for equispaced and unequispaced data
8. Evaluation of definite integrals using Trapezoidal, Simpson's (1/3)rd and (3/8)th rule
9. Gauss-Seidel iteration method to solve the system of linear equations
10. Finding the largest eigen value and the corresponding eigen vector using Rayleigh power method.

Course Assessment and Evaluation:

Continuous Internal Evaluation (CIE): 50 Marks		
Assessment tool	Marks	Course outcomes attained
Internal Test-I	30	CO1, CO2, CO3
Internal Test-II	30	CO3, CO4, CO5
Average of the two tests shall be taken		
Lab components		
Conducting Experiment and Laboratory Record	10	CO1, CO2, CO3, CO4, CO5
Lab Test	10	CO1, CO2, CO3, CO4, CO5
Semester End Examination (SEE)		
Course end examination (Answer any one question from each unit – Internal choice)	100	CO1, CO2, CO3, CO4, CO5